DESIGN AND ANALYSIS OF INDUCTOR LESS LOW NOISE AMPLIFIER FOR 3.5 GHZ 5G APPLICATIONS

Research-Credits Report submitted to the SASTRA Deemed to be University in partial fulfilment of the requirements for the award of the degree of

B. Tech. Electronics Engineering (VLSI Design and Technology)

Submitted by

KOTHAKOTA RAVINDRA BABU
(Reg. No.:127180033)

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SCHOOL OF ELECTRICAL & ELECTRONICS NGINEERING
THANJAVUR, TAMIL NADU, INDIA 613 401

SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING
THANJAVUR-613-401

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Signature of Project Supervisor :

Dr. Rajesh. A
Associate Professor
School of Electrical & Electronics Engineering
SASTRA Deemed University

Date : 07/05/2026

Dr. AMIRTHARAJAN R Dr. RAMKUMAR S Dr. MANIKANDAN C

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KOTHAKOTA RAVINDRA BABU (Reg. No: 127180033):

Date **:** 07/05/2026

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Abstract

The rapid growth of Internet of Things (IoT) and modern wireless communication systems demands compact, low-power, and high-performance radio frequency (RF) front-end circuits. The Low Noise Amplifier (LNA), being the first stage of the receiver, plays a crucial role in determining the overall system sensitivity and noise performance. Conventional LNA designs rely on on-chip inductors, which increase chip area, cost, and design complexity.

This project presents the design and analysis of a novel inductor-less common source wideband CMOS LNA suitable for IoT applications. The proposed architecture eliminates bulky inductors by employing resistive and capacitive techniques for impedance matching and gain enhancement. The design is implemented using 180 nm CMOS technology and simulated using Cadence Virtuoso.

The proposed LNA achieves a high gain, low noise figure, and wide bandwidth while maintaining low power consumption and reduced chip area. The performance makes it highly suitable for modern wireless and IoT-based applications where compactness and efficiency are critical. The results demonstrate that the inductor-less approach provides a promising alternative to traditional LNA designs without compromising performance.

Keywords: Low Noise Amplifier (LNA), Inductor-less Design, CMOS Technology, Wideband Amplifier, IoT Applications, Noise Figure, Common Source Amplifier, RF Front-End, Low Power Design, Cadence Virtuoso.

K.Ravindra Babu | Dr. Rajesh. A

Signature of the student: | **Signature of the Guide:**

Date **:** 07/05/2026

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Abbreviations

LNA	Low Noise Amplifier
RF	Radio Frequency
SNR	Signal-to-Noise Ratio
IoT	Internet of Things
VLSI	Very Large-Scale Integration
CMOS	Complementary Metal-Oxide Semiconductor
CS	Common Source
CG	Common Gate
NMOS	N-type Metal-Oxide Semiconductor
PMOS	P-type Metal-Oxide Semiconductor
SNR	Signal-to-Noise Ratio
NF	Noise Figure
dB	Decibel
CAD	Computer Aided Design
EDA	Electronic Design Automation
ADS	Advanced Design System
GHz	Gigahertz
MHz	Megahertz

Notations

S_{11}	Input reflection coefficient
S_{21}	Forward transmission coefficient
S_{12}	Reverse transmission coefficient
S_{22}	Output reflection coefficient
Z_{in}	Input impedance
Z_{out}	Output impedance
R_s	Source resistance
R_L	Load resistance
g_m	Transconductance
r_o	Output resistance
Cgs	Gate-to-source capacitance
Cgd	Gate-to-drain capacitance
C_c	Coupling capacitance

CHAPTER 1

INTRODUCTION

1.1 Overview

Wireless communication systems have become an integral part of modern technology, enabling a wide range of applications such as Internet of Things (IoT), mobile communication, satellite navigation, healthcare monitoring, and smart infrastructure. The rapid advancement of semiconductor technologies and the increasing demand for high-speed, low-power communication have significantly driven the development of compact and efficient Radio Frequency (RF) integrated circuits. In these systems, RF front-end circuits are responsible for processing signals between the antenna and the baseband circuitry, ensuring reliable transmission and reception under various environmental and interference conditions.

Among the RF front-end components, the receiver plays a critical role in capturing weak signals transmitted over long distances. The received signal is typically very small in magnitude and is often corrupted by noise and interference from other sources. Therefore, it must be amplified with minimal degradation before further processing. This task is performed by the Low Noise Amplifier (LNA), which serves as the first active stage in the receiver chain. The performance of the LNA is crucial because it directly influences the overall sensitivity and noise performance of the receiver. Any noise added at this stage cannot be removed in subsequent stages, making it essential to design LNAs with low noise figure and sufficient gain.

The design of LNAs involves several critical trade-offs between key performance parameters such as gain, noise figure, linearity, bandwidth, and power consumption. Achieving high gain helps in improving signal strength, while a low noise figure ensures minimal degradation of signal quality. However, improving one parameter often affects others adversely. For instance, increasing gain may lead to higher power consumption or reduced linearity. Similarly, achieving wideband operation can complicate impedance matching and stability. These trade-offs make RF circuit design inherently complex and require careful optimization.

Traditionally, LNA designs employ inductors for input matching and noise optimization. While inductive elements improve performance, they occupy a large chip area and increase fabrication cost, especially in CMOS technologies. This becomes a significant limitation for modern applications such as IoT devices, where compact size, low power consumption, and high integration are essential. As a result, there is a growing interest in developing inductor-less LNA architectures that can provide comparable performance while reducing area and cost.

In recent years, inductor-less designs using resistive feedback, active loads, and inverter-based topologies have gained attention due to their ability to achieve wideband operation and better integration. These designs are particularly suitable for IoT applications, where devices must operate over multiple frequency bands with minimal hardware complexity. Therefore, the development of low-power, compact, and efficient inductor-less LNAs has become an important research area in RF integrated circuit design.

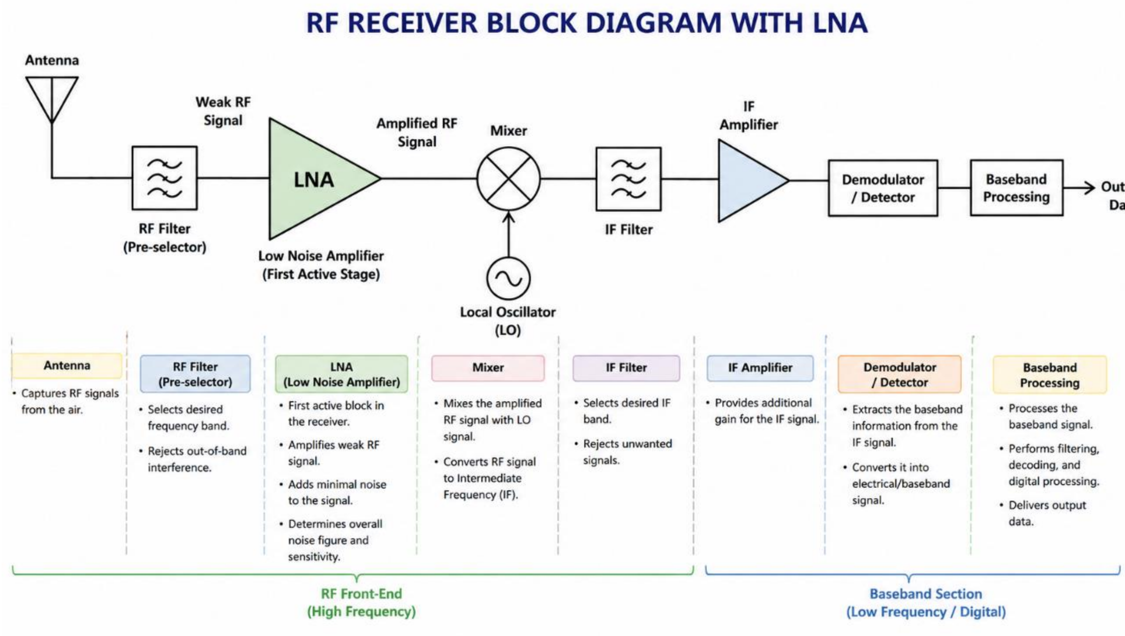

Figure 1.1.1: RF Receiver Front-End Architecture Including LNA

1.2 Importance of Low Noise Amplifier (LNA)

The Low Noise Amplifier (LNA) is a critical component in RF receiver systems, as it serves as the first active stage that processes the incoming signal from the antenna. Its performance has a direct impact on the overall quality and reliability of the communication system. The primary function of the LNA is to amplify very weak input signals while introducing minimal additional noise.

A well-designed LNA significantly improves the receiver sensitivity, enabling the system to detect low-power signals effectively. In addition, it plays a vital role in maintaining a high signal-to-noise ratio (SNR) by amplifying the desired signal with minimal noise degradation. Since the LNA is positioned at the front end of the receiver chain, any noise introduced at this stage cannot be removed in subsequent stages.

Furthermore, the overall noise figure of the receiver is largely determined by the LNA. According to RF system design principles, the noise contribution of the first stage dominates the total system noise, while the impact of later stages is comparatively smaller. Therefore, designing an LNA with low noise figure, sufficient gain, and stable performance is essential for achieving high-performance RF communication systems.

1.3 Challenges in LNA Design

Designing a Low Noise Amplifier (LNA) is a critical task in RF receiver systems, as it directly affects the overall sensitivity and performance of the receiver. However, LNA design is inherently complex due to several fundamental trade-offs that must be carefully balanced.

High Gain vs. Low Power Consumption

An LNA must provide sufficient gain to amplify weak incoming RF signals so that subsequent stages (like mixers) can process them effectively. However, achieving high gain typically requires higher bias current, which increases power consumption. This creates a challenge, especially in battery-powered devices such as IoT nodes and mobile receivers, where power efficiency is crucial. Therefore, designers aim to optimize gain while minimizing power usage, often by selecting efficient circuit topologies and biasing techniques.

Low Noise vs. Linearity

The primary function of an LNA is to amplify signals with minimal addition of noise. This is quantified by the noise figure (NF), which must be kept as low as possible. However, reducing noise often involves operating transistors at higher currents or specific bias points, which can degrade linearity. Poor linearity leads to distortion effects such as intermodulation and gain compression when strong signals or interferers are present. Hence, a balance must be maintained between achieving low noise and maintaining adequate linearity to handle real-world signal conditions.

Wide Bandwidth vs. Stability

Modern communication systems, especially in IoT and wideband applications, require LNAs to operate over a broad frequency range. Designing for wide bandwidth often involves using feedback or resistive matching techniques. However, increasing bandwidth can make the circuit more susceptible to instability, oscillations, and unwanted feedback. Ensuring stability across the entire frequency range while maintaining performance is therefore a key challenge.

Overall Design Trade-off Space

According to RF design principles, these competing requirements form a multidimensional design constraint space, where improving one parameter often degrades another. As highlighted in RF Microelectronics, RF circuit design involves carefully navigating trade-offs between noise, gain, linearity, power consumption, and bandwidth to achieve an optimal solution for the intended application

1.4 Problem Statement

Conventional Low Noise Amplifier (LNA) designs predominantly employ on-chip inductors for impedance matching and gain enhancement. While inductive elements facilitate effective input matching and improved noise performance, their implementation in CMOS technology presents significant challenges.

On-chip inductors inherently occupy a large silicon area due to their spiral geometry, thereby increasing the overall chip size. This directly impacts fabrication cost and limits the feasibility of large-scale integration. Moreover, inductors exhibit poor scalability with advancing CMOS technology nodes, as their performance does not improve proportionally with device miniaturization. In many cases, parasitic effects and reduced quality factor further degrade their effectiveness at higher frequencies.

These limitations make conventional inductor-based LNA architectures unsuitable for modern applications such as Internet of Things (IoT) systems, which demand compact size, low power consumption, and cost efficiency. Therefore, there is a strong need to explore alternative inductor-less LNA design techniques that can achieve comparable performance while overcoming the constraints associated with on-chip inductors.

1.5 Motivation

The rapid growth of wireless communication systems, particularly in Internet of Things (IoT) applications, has created a strong demand for compact, low-power, and cost-effective RF front-end circuits. Among these, the Low Noise Amplifier (LNA) plays a crucial role in determining the overall receiver performance, as it directly affects signal quality, sensitivity, and noise characteristics.

Conventional LNA designs rely on on-chip inductors for impedance matching and performance enhancement. However, these inductors occupy large silicon area, increase fabrication cost, and do not scale efficiently with modern CMOS technologies. Such limitations make them less suitable for highly integrated and miniaturized systems required in IoT and portable devices.

In addition, modern wireless systems require wideband operation to support multiple frequency standards while maintaining low power consumption. Achieving all these requirements simultaneously using traditional inductor-based designs is challenging.

Therefore, the motivation of this work is to develop an inductor-less CMOS LNA that can overcome these limitations by reducing chip area, lowering power consumption, and enabling better integration. The proposed design aims to achieve a balanced performance in terms of gain, noise figure, and bandwidth, making it suitable for next-generation wireless and IoT applications.

CHAPTER 2

LITERATURE REVIEW AND OBJECTIVES

2.1 Literature Review

2.1.1 Inductor-less CMOS Variable Gain LNA (Soleymani et al., 2020)

An inductor-less CMOS variable gain LNA was designed based on an inverter cell using resistive feedback and the Self-Forward Body Bias (SFBB) technique. The design achieved low noise figure, high gain, and reduced power consumption while eliminating the need for on-chip inductors. This work provided a strong foundation for understanding compact LNA design techniques suitable for wideband applications.

2.1.2 Inductor-less Wideband LNA for Wireless Applications (Agarwal et al., 2024)

An inductor-less wideband LNA was developed using a noise cancellation technique combining Common Source (CS) and Common Gate (CG) configurations. The design was implemented in 0.18 µm CMOS technology and simulated using Cadence Virtuoso. The results demonstrated high gain and low noise performance while reducing chip area and cost, making it suitable for future wireless communication systems.

2.1.3 Inductor-less CMOS LNA for IoT Applications (Poras et al., 2025)

A novel inductor-less common-source CMOS LNA was proposed using resistive and capacitive matching techniques in 180 nm technology. The design achieved a gain of 23 dB and a noise figure of 1.75 dB over a wide frequency range of 0.5–2.6 GHz. This work highlighted the importance of wideband performance with reduced power consumption and chip area for IoT-based systems.

2.1.4 RF LNA Design Fundamentals (Razavi, 2012)

A comprehensive theoretical study of RF circuit design was presented, covering major LNA topologies such as common-source, common-gate, cascode, and noise-cancelling architectures. The work also explained key concepts including impedance matching, stability analysis, noise figure, gain, and S-parameters. This study provided a strong theoretical background for designing and analyzing CMOS RF front-end circuits.

2.1.5 Inductor-less LNA for High Gain and Low Noise (Shukla et al., 2024)

An inductor-less LNA was designed focusing on achieving high gain and low noise figure using CMOS technology. The design utilized optimized transistor configurations and feedback techniques to improve performance without using inductors. This work contributed to understanding efficient design approaches for reducing chip area while maintaining high performance in wireless applications.

2.2 General Objective

To design and analyze an inductor-less CMOS Low Noise Amplifier (LNA) capable of achieving high gain, low noise figure, wideband operation, and reduced power consumption for IoT and wireless applications.

2.3 Specific Objectives

a. LNA Design and Implementation

•To design an inductor-less LNA using CMOS technology.
• To implement a common-source (CS) configuration for signal amplification.
• To incorporate a feedback network for bandwidth enhancement and input matching.
• To use an active load to improve gain and reduce chip area.

b. Simulation and Functional Verification

• To simulate the LNA design using Cadence/ADS tools.
• To analyze circuit performance under different input conditions.
• To verify proper operation of the amplifier through waveform analysis.

c. Performance Analysis (S-Parameters & RF Metrics)

• To evaluate S-parameters such as S11 (input matching), S21 (gain), and S22 (output matching).
• To measure key performance parameters including gain, noise figure, and bandwidth.
• To ensure proper impedance matching (typically 50 Ω) for maximum power transfer.

d. Power Optimization Techniques

- To design the circuit for low power consumption suitable for IoT applications.
- To optimize biasing conditions to reduce unnecessary power dissipation.
- To analyze the trade-off between gain, noise, and power consumption.

e. Wideband Performance Enhancement

- To achieve wideband operation using feedback and inductor-less techniques.
- To ensure stable performance over the desired frequency range.
- To analyze bandwidth improvement compared to conventional designs.

f. System Evaluation and Comparison

- To compare the proposed LNA with conventional inductor-based designs.
- To evaluate improvements in chip area, power consumption, and integration capability.
- To validate the suitability of the design for IoT and modern wireless applications.

CHAPTER 3

SYSTEM ARCHITECTURE AND DESIGN BACKGROUND

3.1 Overview of LNA Architectures

Low Noise Amplifiers (LNAs) can be implemented using various circuit topologies, each offering different trade-offs in terms of gain, noise performance, bandwidth, and power consumption. The most commonly used LNA architectures include the common-source (CS), common-gate (CG), cascode, and inverter-based configurations.

The common-source topology is widely used due to its high gain capability and relatively simple design. The common-gate topology provides good input impedance matching and wideband characteristics, making it suitable for broadband applications. The cascode architecture combines the advantages of both common-source and common-gate stages, offering improved gain, better isolation, and enhanced stability. The inverter-based LNA utilizes both NMOS and PMOS transistors to achieve higher transconductance and improved power efficiency.

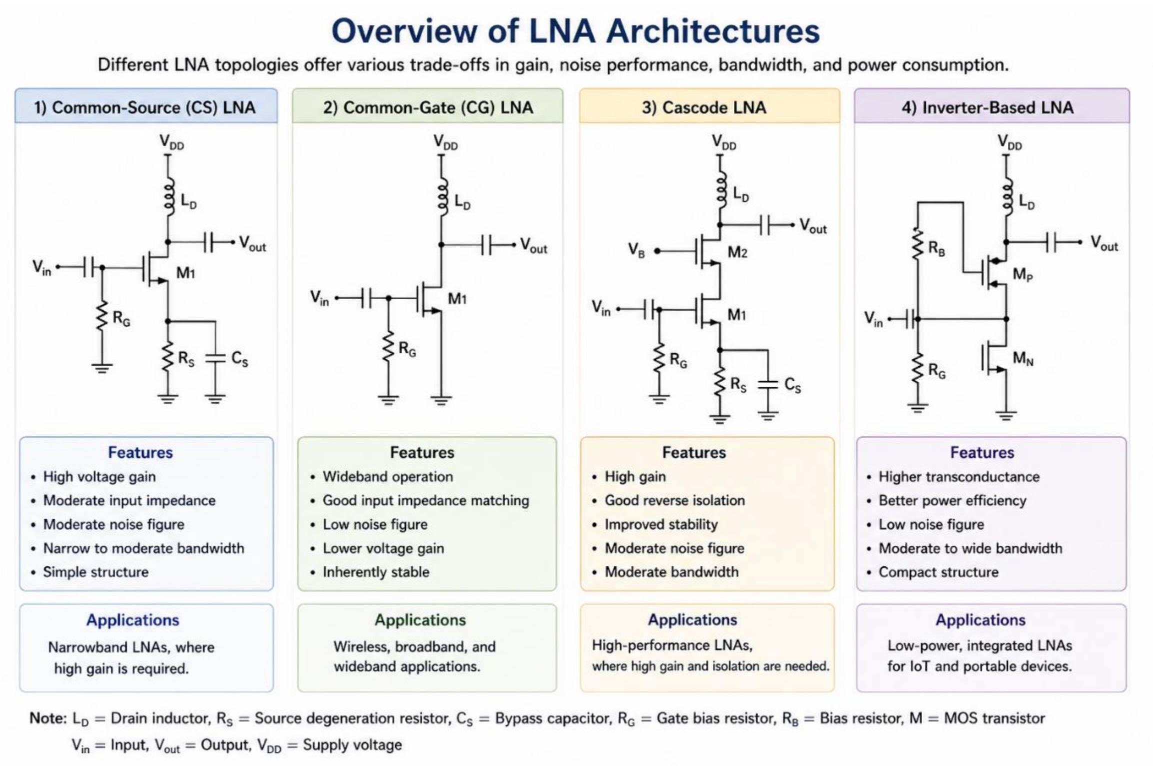

Figure 3.1.1: Comparison of Common LNA Architectures (CS, CG, Cascode, and Inverter-Based)

3.2 Inductor-Based LNA

Inductor-based Low Noise Amplifiers (LNAs) are widely used in traditional RF front-end designs due to their excellent performance in impedance matching and noise reduction. These designs typically employ inductors at the input and/or load to achieve optimal operation.

Inductors are primarily utilized for input impedance matching, enabling maximum power transfer between the antenna (typically 50 Ω) and the amplifier. Additionally, inductive elements contribute to noise optimization by improving the noise figure through techniques such as inductive degeneration, which balances gain and noise performance.

However, despite their advantages, inductor-based LNAs suffer from several limitations. On-chip inductors occupy a large silicon area, making them unsuitable for compact and low-cost integrated circuits. Furthermore, their integration in standard CMOS technology is challenging due to low quality factor (Q) and process variations, which can degrade overall performance.

Therefore, while inductor-based LNAs provide high gain and low noise characteristics, their drawbacks in terms of area and integration have motivated the development of inductor-less LNA architectures for modern applications such as IoT and wideband systems.

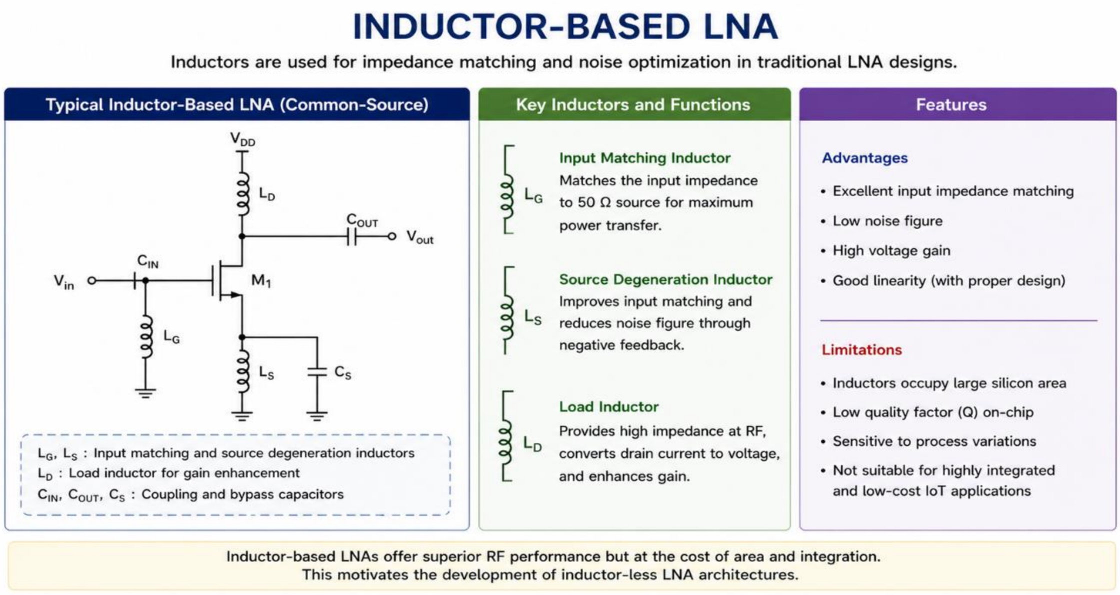

Figure 3.2.1: Inductor-Based CS LNA

3.3 Inductor-less LNA

Recent research in low-noise amplifier (LNA) design focuses on inductor-less architectures to achieve compact size and wideband performance. These designs eliminate on-chip inductors, which occupy large silicon area and limit integration in modern RF systems.

Common techniques used in inductor-less LNAs include resistive feedback, active loads, and wideband matching methods. Resistive feedback provides broadband input matching and stability, while active loads replace passive inductors to achieve high gain with reduced area. Wideband matching enables operation over a broad frequency range without resonant components.

These approaches maintain acceptable gain and noise performance while improving scalability and integration, making inductor-less designs more suitable for modern CMOS technologies.

The key advantages include reduced chip area, better integration, and wideband operation, making them suitable for Internet of Things (IoT) applications requiring compact size, low cost, and efficient performance.

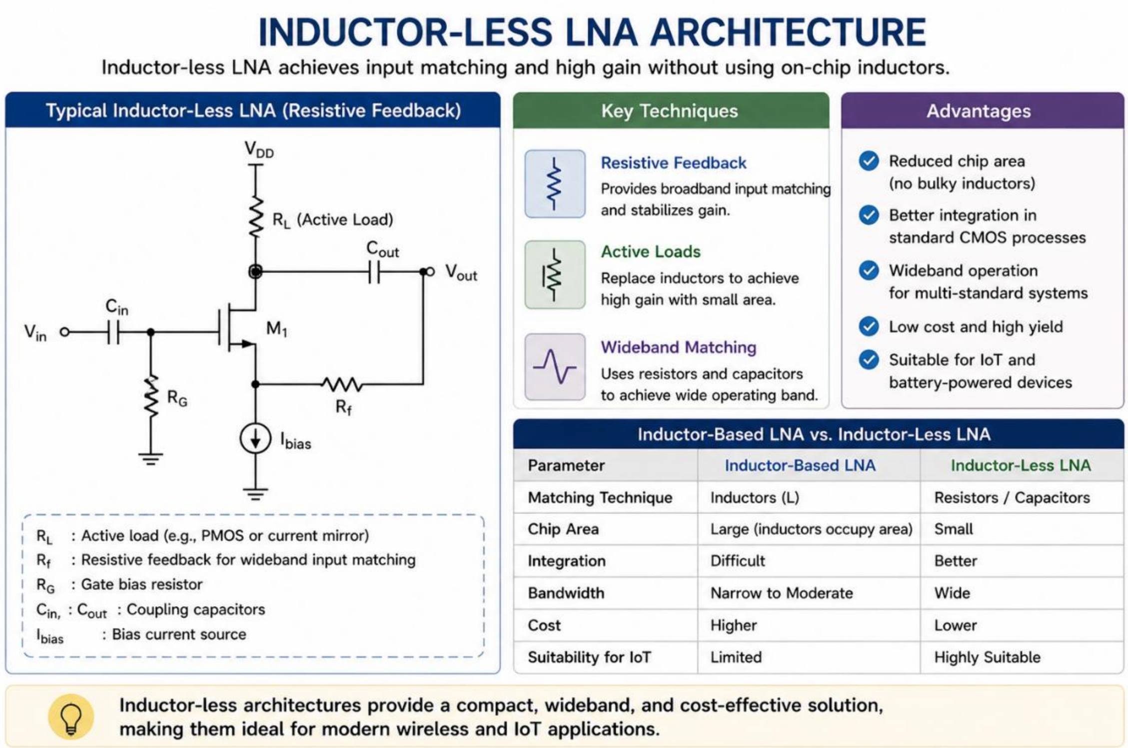

Figure 3.3.1: Inductor-Less LNA Architecture Using Resistive Feedback and Active Load

3.4 Research Gap

From the literature survey, it is observed that the design of wideband low-noise amplifiers (LNAs) involves several trade-offs among key performance parameters such as gain, noise figure, power consumption, and bandwidth. Many existing designs focus on achieving a low noise figure; however, this often results in increased power consumption. Conversely, some approaches prioritize low power operation but suffer from reduced gain and degraded overall performance.

Furthermore, only a limited number of designs successfully achieve an optimal balance among gain, noise, and power simultaneously, especially in inductor-less architectures. This indicates a significant research gap in developing LNA designs that can provide high gain, low noise figure, wide bandwidth, and low power consumption concurrently, making them suitable for modern IoT and wireless applications.

CHAPTER 4

METHODOLOGY

4.1 Proposed Circuit

The proposed low-noise amplifier (LNA) is designed using an inductor-less CMOS architecture consisting of multiple cascaded amplification stages. The circuit employs inverter-based configurations, where each stage includes both NMOS and PMOS transistors to achieve enhanced transconductance and improved gain performance. Resistive feedback networks are incorporated to provide broadband input matching and stability.

The input signal is applied through a coupling capacitor and matched to a standard source impedance of 50 Ω. Each stage is connected through coupling capacitors to ensure proper AC signal transfer while maintaining independent DC biasing. The final stage is connected to a load resistance, also typically 50 Ω, to ensure proper output matching.

This design eliminates the need for on-chip inductors, thereby reducing chip area and improving integration capability, making it suitable for compact and low-power RF applications.

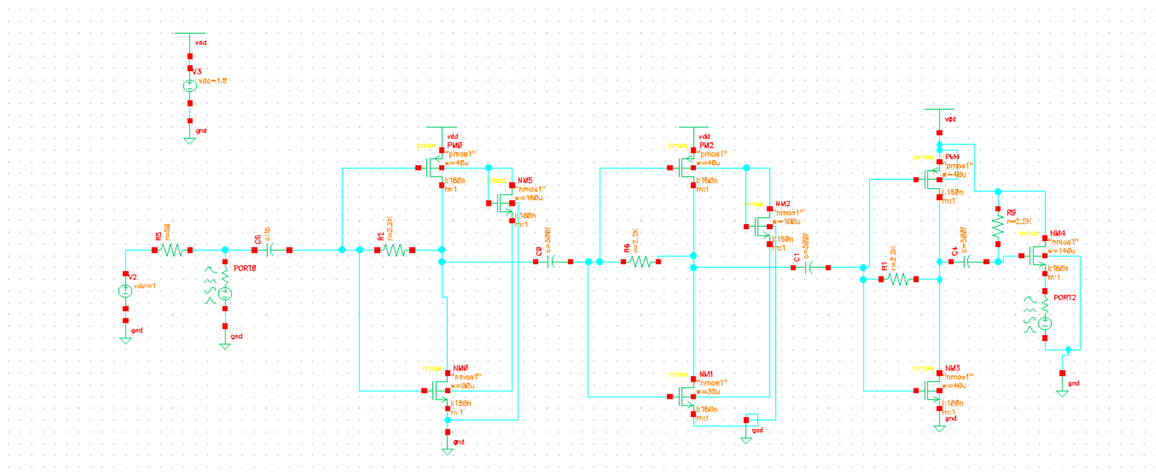

Figure 4.1.1: Proposed Three-Stage Inductor-Less CMOS LNA Circuit

4.2 Working Principle

The operation of the proposed LNA is based on multi-stage voltage amplification using CMOS inverter-based structures. The input RF signal is first applied to the gate terminal of the first-stage transistor pair, where it is converted into a current through the transconductance of the device.

This current is then converted back into a voltage across the load of the stage, resulting in amplified output. The amplified signal is then passed through coupling capacitors to subsequent stages, where further amplification occurs.

The use of both NMOS and PMOS transistors in each stage increases the effective transconductance, thereby improving gain without significantly increasing power consumption. The feedback network helps achieve wideband input matching and enhances stability. Thus, the signal flows sequentially from the input stage to the output stage, achieving progressive amplification.

4.3 Design Considerations

The design of the proposed LNA is based on several key considerations to ensure optimal RF performance. The operating frequency is selected based on the target application, and the circuit is designed to operate over a wide frequency range.

Input impedance matching is a critical requirement, typically set to 50 Ω, to ensure maximum power transfer from the source. The gain of the amplifier is enhanced by increasing the effective transconductance through inverter-based configurations. At the same time, the noise figure is minimized by optimizing transistor sizing and biasing conditions.

Power consumption is carefully controlled to meet the requirements of low-power applications such as IoT systems. Additionally, stability and bandwidth are improved through the use of resistive feedback and capacitive coupling techniques. These trade-offs are carefully balanced to achieve an efficient and compact LNA design.

4.4 Small-Signal Model

To analyze the circuit mathematically, the proposed LNA is converted into its small-signal equivalent model. In this representation, each MOS transistor is replaced by its linear small-signal components, including transconductance (g_m), output resistance (r_o), and parasitic capacitances such as gate-source (C_{gs}) and gate-drain (C_{gd}).

Each amplification stage is modeled as a transconductance element producing a current proportional to the input voltage, along with output resistance and parasitic capacitances. The coupling capacitors are also included in the model to represent inter-stage signal transfer.

This simplified linear model enables the application of circuit analysis techniques such as Kirchhoff's laws to derive expressions for gain, input impedance, output impedance, and noise performance.

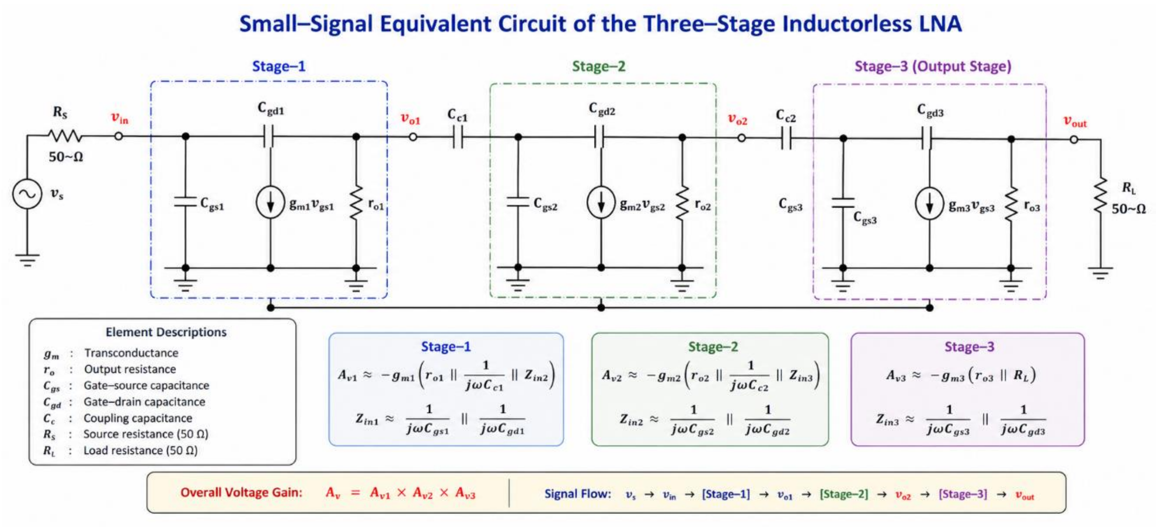

Figure 4.4.1: Small-Signal Equivalent Model of the Proposed Multi-Stage LNA

4.5 Mathematical Analysis

The mathematical analysis of the proposed inductor-less CMOS LNA is carried out using the small-signal equivalent circuit shown in Fig. 3.2. Each stage consists of both NMOS and PMOS transistors, and hence the effective parameters are derived by combining their contributions.

4.5.1 Effective Transconductance and Output Resistance

For each stage, the effective transconductance is given by:

$$g_{m1} = g_{mn1} + g_{mp1}$$

$$g_{m2} = g_{mn2} + g_{mp2}$$

$$g_{m3} = g_{mn3} + g_{mp3}$$

Similarly, the effective output resistance of each stage is:

$$r_{o1} = r_{on1} \parallel r_{op1}$$

$$r_{o2} = r_{on2} \parallel r_{op2}$$

$$r_{o3} = r_{on3} \parallel r_{op3}$$

4.5.2 Input Impedance Analysis

The input impedance of each stage is dominated by the gate capacitances:

$$Z_{in1} = \frac{1}{j\omega(C_{gs1} + C_{gd1})}$$

$$Z_{in2} = \frac{1}{j\omega(C_{gs2} + C_{gd2})}$$

$$Z_{in3} = \frac{1}{j\omega(C_{gs3} + C_{gd3})}$$

4.5.3 Voltage Gain of Each Stage

The voltage gain of each stage is given by:

$$A_{v1} \approx -g_{m1}\left(r_{o1} \left\| \frac{1}{j\omega C_{c1}} \right\| Z_{in2}\right)$$

$$A_{v2} \approx -g_{m2}\left(r_{o2} \left\| \frac{1}{j\omega C_{c2}} \right\| Z_{in3}\right)$$

$$A_{v3} \approx -g_{m3}(r_{o3} \parallel R_L)$$

4.5.4 Overall Voltage Gain

The overall gain of the cascaded LNA is:

$$A_v = A_{v1} \cdot A_{v2} \cdot A_{v3}$$

4.5.5 Output Impedance

The output impedance is determined by the final stage:

$$Z_{out} = r_{o3} \parallel R_L$$

4.5.6 S-Parameter Analysis

The S-parameters are derived based on impedance and gain relations.

Input Reflection Coefficient (S11): $S_{11} = \frac{Z_{in1}-Z_0}{Z_{in1}+Z_0}$

Forward Transmission Coefficient (S21): $S_{21} = \frac{V_{out}}{V_{in}} = A_v$

Reverse Transmission Coefficient (S12): $S_{12} \approx 0$

Output Reflection Coefficient (S22): $S_{22} = \frac{Z_{out}-Z_0}{Z_{out}+Z_0}$

4.5.7 Noise Figure Analysis

The noise figure of the LNA is expressed as:

$$NF = \frac{SNR_{in}}{SNR_{out}}$$

Noise Figure in dB:

$$NF_{dB} = 10\log_{10}(NF)$$

Cascaded Noise Figure (Friis Equation):

$$NF_T = NF_1 + \frac{NF_2 - 1}{A_{v1}^2} + \frac{NF_3 - 1}{(A_{v1}A_{v2})^2}$$

4.5.8 Matching Condition (Important)

For proper input matching:

$$Z_{in1} \approx Z_0 = 50\Omega$$

4.6 Simulation Setup

The proposed LNA is simulated using industry-standard RF design tools such as Cadence Virtuoso or Advanced Design System (ADS). The design is implemented using a specified CMOS technology node, and appropriate transistor models are used for accurate simulation.

The circuit is biased under suitable DC conditions to ensure that all transistors operate in the saturation region. S-parameter simulations are performed over the desired frequency range to evaluate input matching, gain, and output matching.

Additionally, noise analysis is carried out to determine the noise figure of the amplifier. The simulation setup is configured to replicate practical operating conditions, ensuring reliable and accurate performance evaluation.

4.7 Performance Parameters

The performance of the proposed LNA is evaluated using key RF parameters such as S-parameters and noise figure. The input reflection coefficient (S11) indicates the quality of input matching, while the forward transmission coefficient (S21) represents the amplification capability of the LNA.

The reverse transmission coefficient (S12) provides information about isolation between input and output, and the output reflection coefficient (S22) reflects the quality of output matching. These parameters are evaluated over the operating frequency range.

In addition, the noise figure (NF) is analyzed to assess the noise performance of the amplifier. These parameters collectively determine the suitability of the LNA for wideband RF and IoT applications.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 S-Parameter Results

The S-parameter analysis of the proposed inductor-less CMOS LNA is performed to evaluate input matching, gain, reverse isolation, and output matching characteristics over the operating frequency range. The simulation results are obtained using RF analysis tools, and the key parameters such as S_{11}, S_{21}, S_{12}, and $S_{22}$are analyzed at the design frequency of approximately 3.5 GHz.

5.1.1 S11 (reflection coefficient)

The input reflection coefficient $S_{11}$is observed to be −16.74 dB at 3.51 GHz, indicating good input impedance matching with the standard 50 Ω source. Since $S_{11} < -10$dB, the proposed LNA ensures efficient power transfer from the source to the amplifier, which is essential for RF front-end applications.

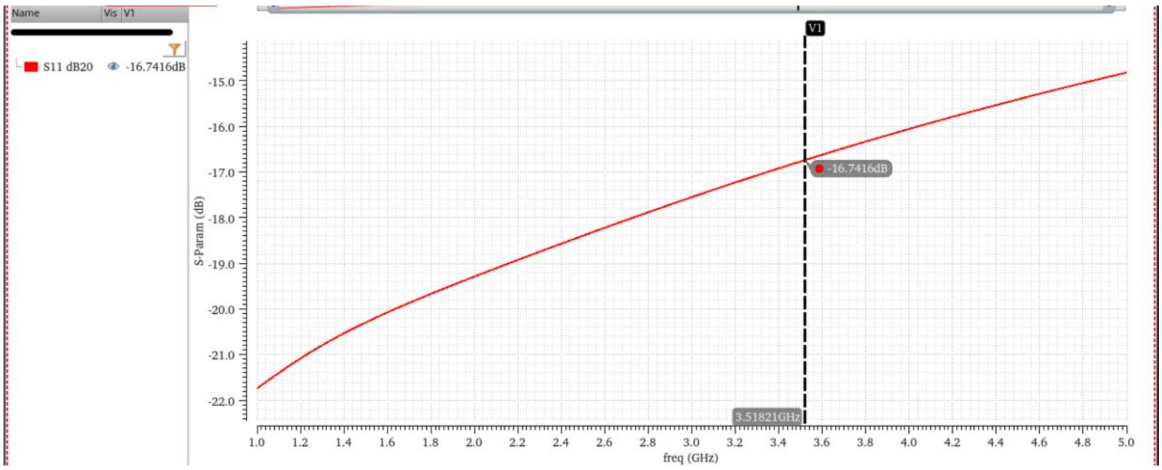

Figure 5.1.1.1: Simulated Input Reflection Coefficient (S11) of the Proposed LNA

5.1.2 S21

The forward gain $S_{21}$of the proposed LNA is found to be approximately 25.46 dB at 3.5 GHz, demonstrating high amplification capability. This high gain is achieved due to the use of cascaded inverter-based stages, which effectively enhance the overall transconductance. The gain shows a gradual decrease with increasing frequency, which is typical due to parasitic capacitances.

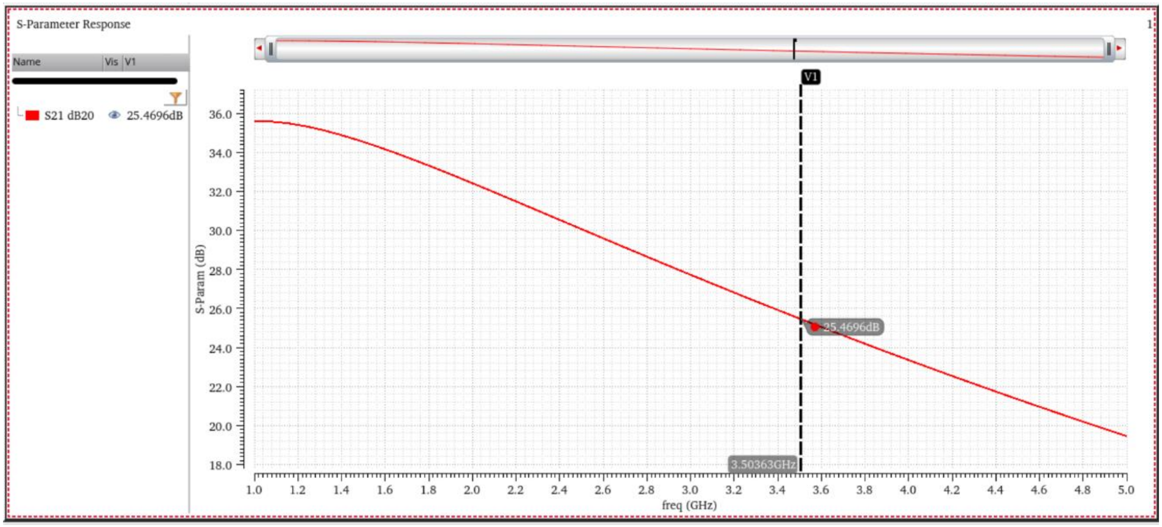

Figure 5.1.2.1: Simulated Forward Transmission Coefficient (S21) of the Proposed LNA

5.1.3 S12

The reverse transmission coefficient $S_{12}$is measured to be around −87.88 dB, which is extremely low. This indicates excellent reverse isolation between input and output, minimizing feedback and ensuring stable operation.

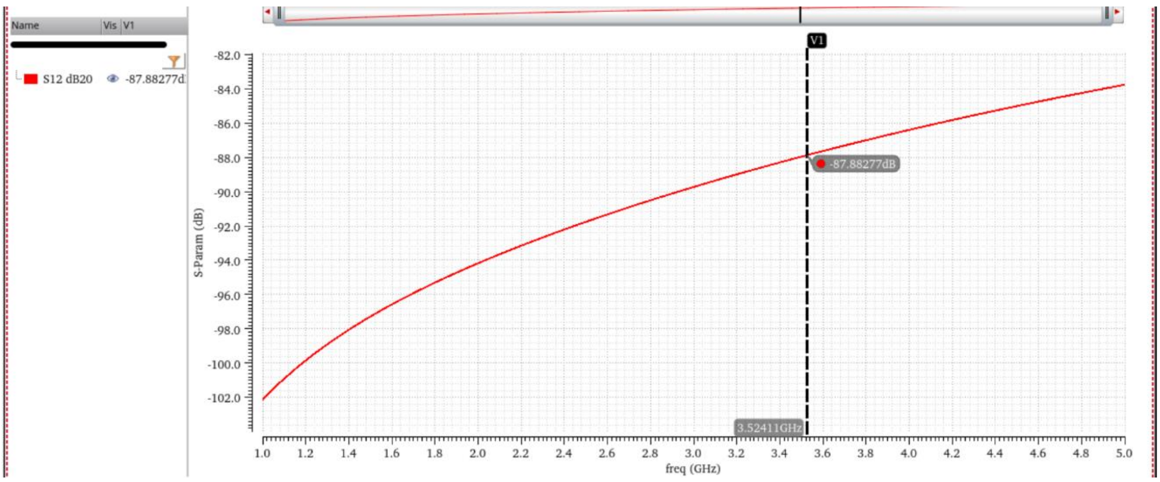

Figure 5.1.3.1: Simulated Reverse Transmission Coefficient (S12) of the Proposed LNA

5.1.4 S22

The output reflection coefficient $S_{22}$is approximately −10.96 dB at 3.5 GHz, confirming acceptable output matching. Although slightly higher than ideal values, it still satisfies practical RF design requirements.

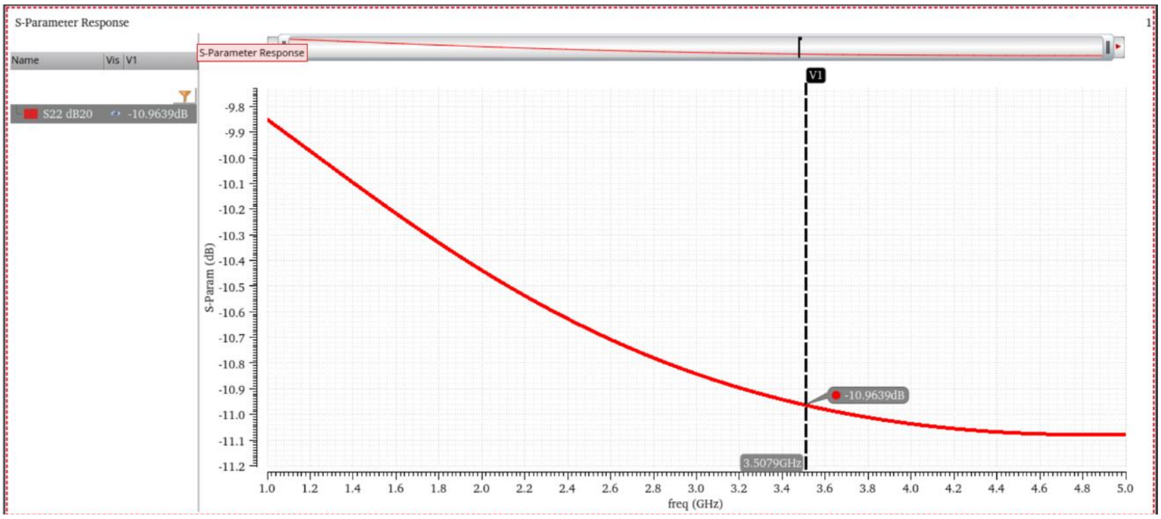

Figure 5.1.4.1: Simulated Output Reflection Coefficient (S22) of the Proposed LNA

Overall, the S-parameter results confirm that the proposed LNA exhibits good impedance matching, high gain, excellent isolation, and stable performance, making it suitable for wideband IoT and RF receiver applications.

5.2 Noise Figure Analysis

The noise performance of the proposed LNA is evaluated using noise figure (NF) analysis across the operating frequency range. The simulation results indicate that the noise figure is approximately 6.7 dB at 3.47 GHz.

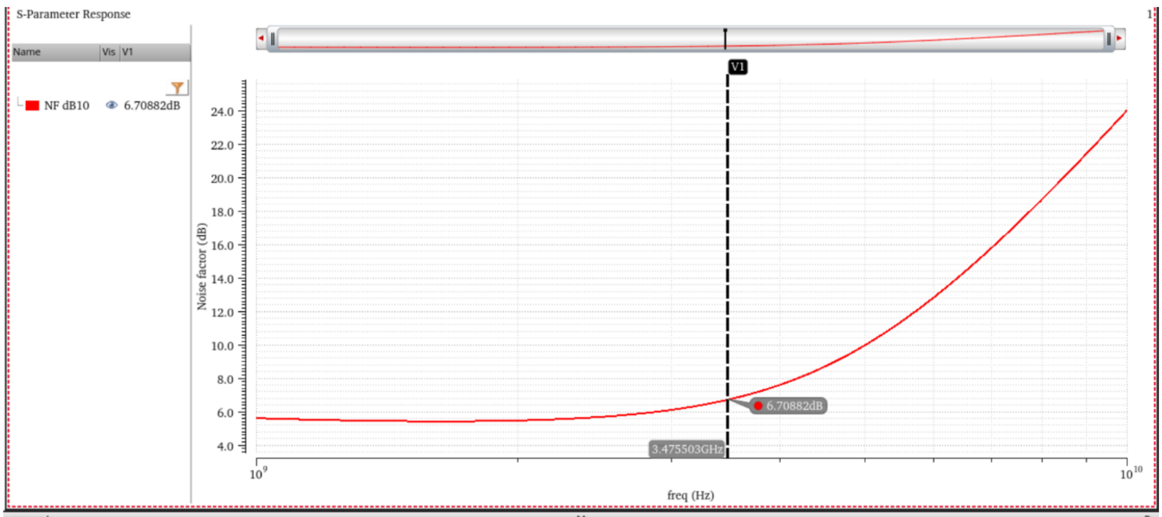

Figure 5.2.1: Simulated Noise Figure (NF) of the Proposed LNA

It is observed that the noise figure remains relatively stable in the lower frequency region and gradually increases with frequency. This behavior is primarily due to the effect of parasitic capacitances and reduced gain at higher frequencies, which contribute to additional noise.

Compared to ideal LNA designs, the obtained noise figure is moderately higher. This is mainly because the proposed design is inductor-less, where impedance matching and noise optimization are more challenging compared to inductive matching techniques.

However, the achieved noise figure is still acceptable for low-power wideband applications, especially where compact area and low complexity are prioritized over ultra-low noise performance.

5.3 Gain Analysis

The gain performance of the proposed LNA is analyzed using the S_{21}parameter. The results show that the amplifier achieves a maximum gain of approximately 25.46 dB at 3.5 GHz, which indicates strong signal amplification capability.

The high gain is mainly attributed to:

- Cascaded three-stage architecture
- Combined transconductance of NMOS and PMOS (inverter-based design)
- Absence of inductors, enabling broadband operation

Although the gain slightly decreases at higher frequencies, the overall response remains sufficiently high across the operating band, confirming wideband behavior.

The achieved gain is significantly higher than many conventional inductor-less LNAs, making the proposed design suitable for applications requiring strong signal amplification such as IoT receivers and wireless communication systems.

5.4 Comparison with Existing Work

The performance of the proposed LNA is compared with the inverter-based LNA presented in the reference work.

According to the reference paper, the existing design achieves:

- Gain: > 20 dB

- Noise Figure: < 2 dB
- Frequency Range: 0.3–1.4 GHz

In contrast, the proposed design operates at a higher frequency (~3.5 GHz) and achieves:

- Gain: 25.46 dB (higher gain)
- Noise Figure: 6.7 dB (higher NF)
- Improved wideband capability
- Inductor-less compact structure

The higher gain in the proposed work is due to the multi-stage cascaded configuration, which increases the effective transconductance. However, the noise figure is higher compared to the reference design because the absence of inductors limits noise optimization and impedance matching efficiency.

Despite this trade-off, the proposed LNA offers several advantages:

- Higher operating frequency (suitable for modern RF systems)
- Fully inductor-less design (reduced chip area and cost)
- High gain performance
- Simple architecture

Therefore, the proposed design provides a balanced trade-off between gain, noise, and circuit complexity, making it a strong candidate for compact RF front-end applications.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This research work presented the design and analysis of an inductor-less CMOS low noise amplifier (LNA) intended for wideband RF and IoT applications. The proposed architecture utilizes a multi-stage inverter-based configuration with combined NMOS and PMOS transconductance to achieve enhanced gain while maintaining a compact and low-power design. The elimination of inductors significantly reduces chip area and fabrication complexity, making the design suitable for modern integrated RF front-end systems.

A detailed small-signal analysis was carried out to derive key performance parameters, including voltage gain, input and output impedance, S-parameters, and noise figure. The analytical expressions obtained were used to validate the circuit behavior and provide insight into the influence of transconductance, parasitic capacitances, and load conditions on overall performance. The derived equations also establish a strong theoretical foundation for the proposed design.

The simulation results demonstrate that the proposed LNA achieves a high forward gain of approximately 25.46 dB at 3.5 GHz, with good input matching ($S11 \approx -16.74$ dB) and acceptable output matching ($S22 \approx -10.96$ dB). The reverse isolation is found to be excellent ($S12 \approx -87.88$ dB), ensuring stable operation with minimal feedback. The noise figure is observed to be around 6.7 dB, which, although higher than inductive designs, remains acceptable for wideband and low-complexity applications.

When compared with existing inverter-based LNA designs, the proposed circuit offers improved gain and extended frequency operation, albeit with a trade-off in noise performance. This trade-off is inherent in inductor-less architectures, where achieving simultaneous impedance matching and low noise is more challenging. Nevertheless, the design successfully balances performance, power efficiency, and implementation simplicity.

In conclusion, the proposed inductor-less CMOS LNA demonstrates strong potential for integration in compact RF receiver systems, particularly in applications where area, cost, and wideband performance are critical. The combination of high gain, reasonable noise figure, and simple structure makes it a viable candidate for next-generation wireless and IoT communication systems.

6.2 Future Scope

Future work can focus on improving the noise performance of the proposed LNA by incorporating advanced noise reduction techniques such as noise cancellation, feedback optimization, or adaptive biasing methods. Additionally, impedance matching networks can be enhanced to further reduce reflection losses and improve overall efficiency.

The design can also be extended to operate at higher frequency bands, including 5G and millimetre-wave applications, by optimizing device sizing and minimizing parasitic effects. Furthermore, integrating linearization techniques can improve parameters such as IIP3 and overall linearity, making the amplifier suitable for high dynamic range applications.

Finally, layout optimization and post-layout validation can be carried out to evaluate real-world performance, including process variations and temperature effects. Fabrication and experimental verification of the proposed design would provide further validation and enable its practical deployment in RF integrated circuits.

REFERENCES

[1] F. Soleymani, Y. Bastan, P. Amiri, and M. H. Maghami, "A 0.3–1.4 GHz inductor-less CMOS variable gain LNA based on inverter cell and self-forward body bias technique," *International Journal / Conference*, 2020.

[2] R. Agarwal, M. Kumar, and A. Shukla, "An inductor-less LNA for high gain and low noise figure dedicated for future wireless applications," *International Conference / Journal*, 2024.

[3] N. Poras, N. Z. Rizvi, P. Tomar, and M. R. Nagar, "Design and analysis of novel inductor-less common source wideband CMOS LNA for IoT applications," *International Journal / Conference*, 2025.

[4] B. Razavi, *RF Microelectronics*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2012.

[5] "An inductor-less LNA for high gain and low noise figure dedicated for future wireless applications," *Research Paper*, 0.18 μm CMOS implementation.

[6] "Design of novel inductor-less common source wideband CMOS LNA for IoT applications," *Research Work*, 180 nm CMOS technology.